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ZHANG et al.(10) **Pub. No.: US 2025/0286387 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **BATTERY DYNAMIC-EQUALIZATION
APPARATUS AND CONTROL METHOD
THEREFOR, AND VEHICLE****B60L 58/22** (2019.01)**B60L 58/27** (2019.01)**H02P 21/22** (2016.01)(71) Applicant: **BYD COMPANY LIMITED,**
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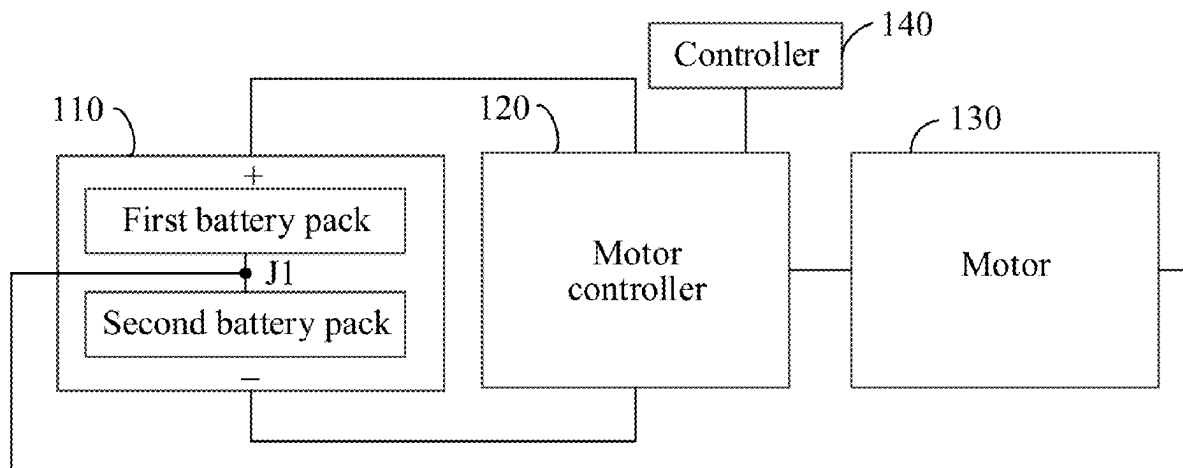
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ABSTRACT(22) Filed: **May 21, 2025****Related U.S. Application Data**(63) Continuation of application No. PCT/CN2023/
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A battery dynamic-equalization apparatus includes a power battery including a first battery pack and a second battery pack, a motor controller connected to the power battery, a motor connected to the motor controller, and a controller connected to the motor controller. The controller is configured to: control, in a first preset state, a bridge arm of the electric-motor controller to drive the electric motor, cause a first battery pack and a second battery pack to be alternately charged and discharged so as to realize self-heating of the first battery pack and the second battery pack, and to configure the absolute value of the difference value between the capacitance of the first battery pack and the capacitance of the second battery pack to be lower than a preset threshold value.



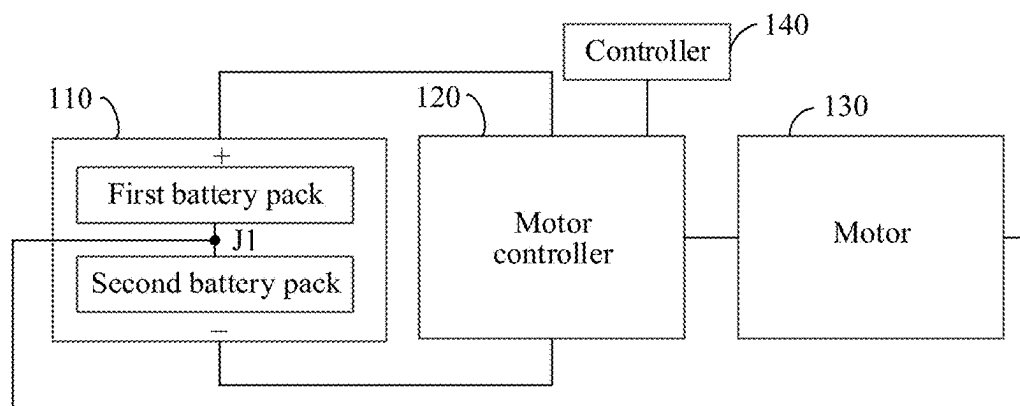


FIG. 1

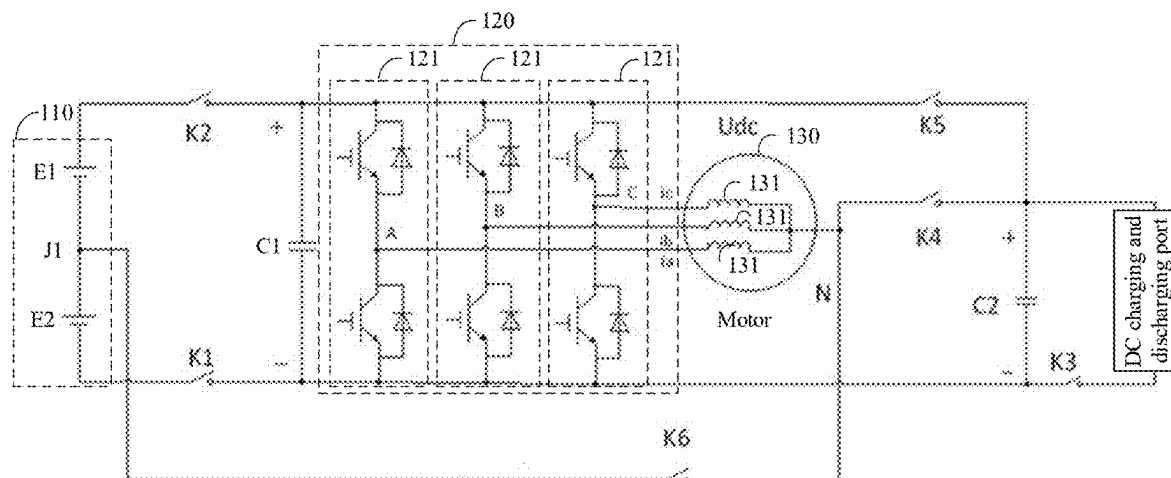


FIG. 2

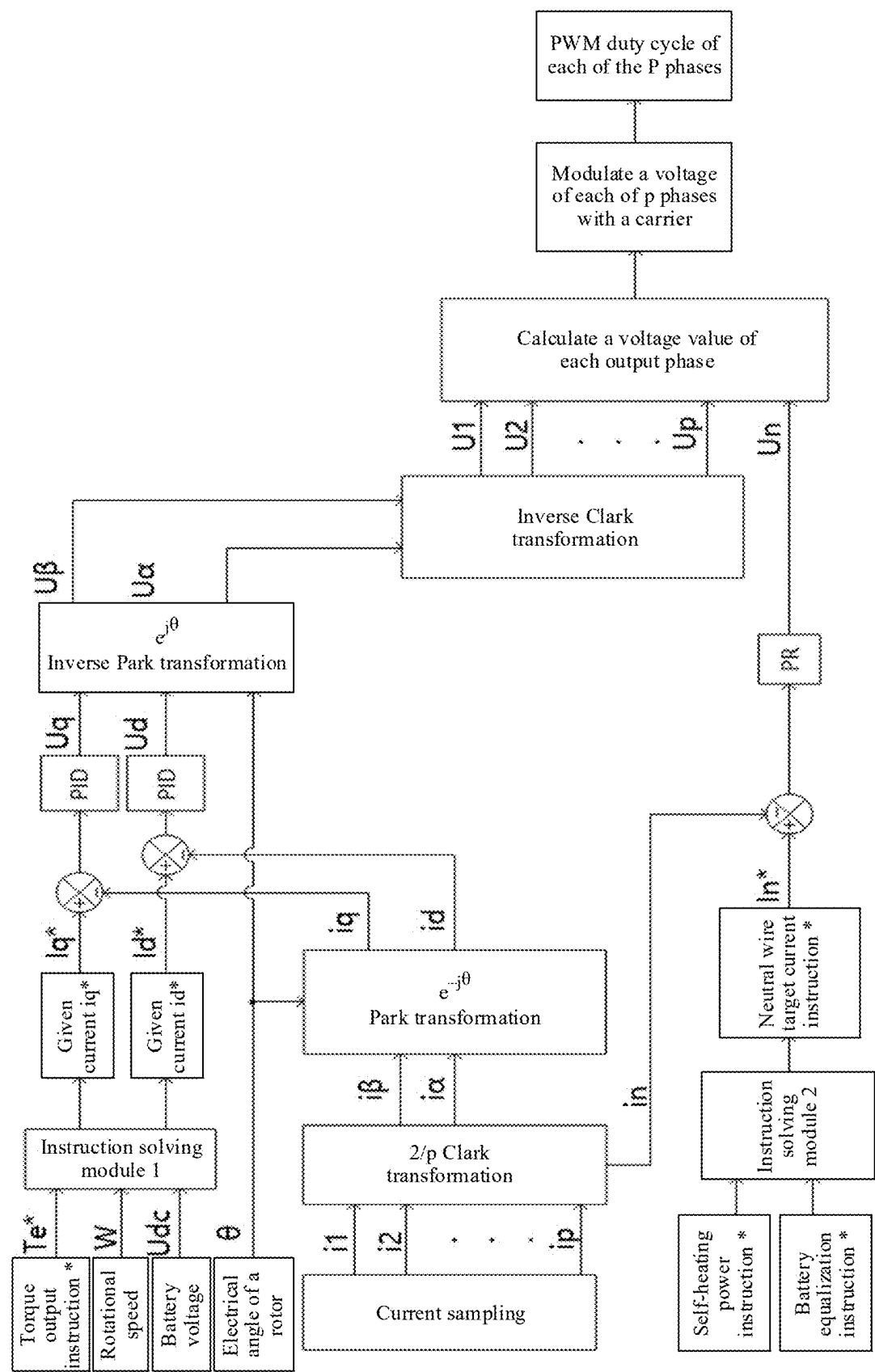
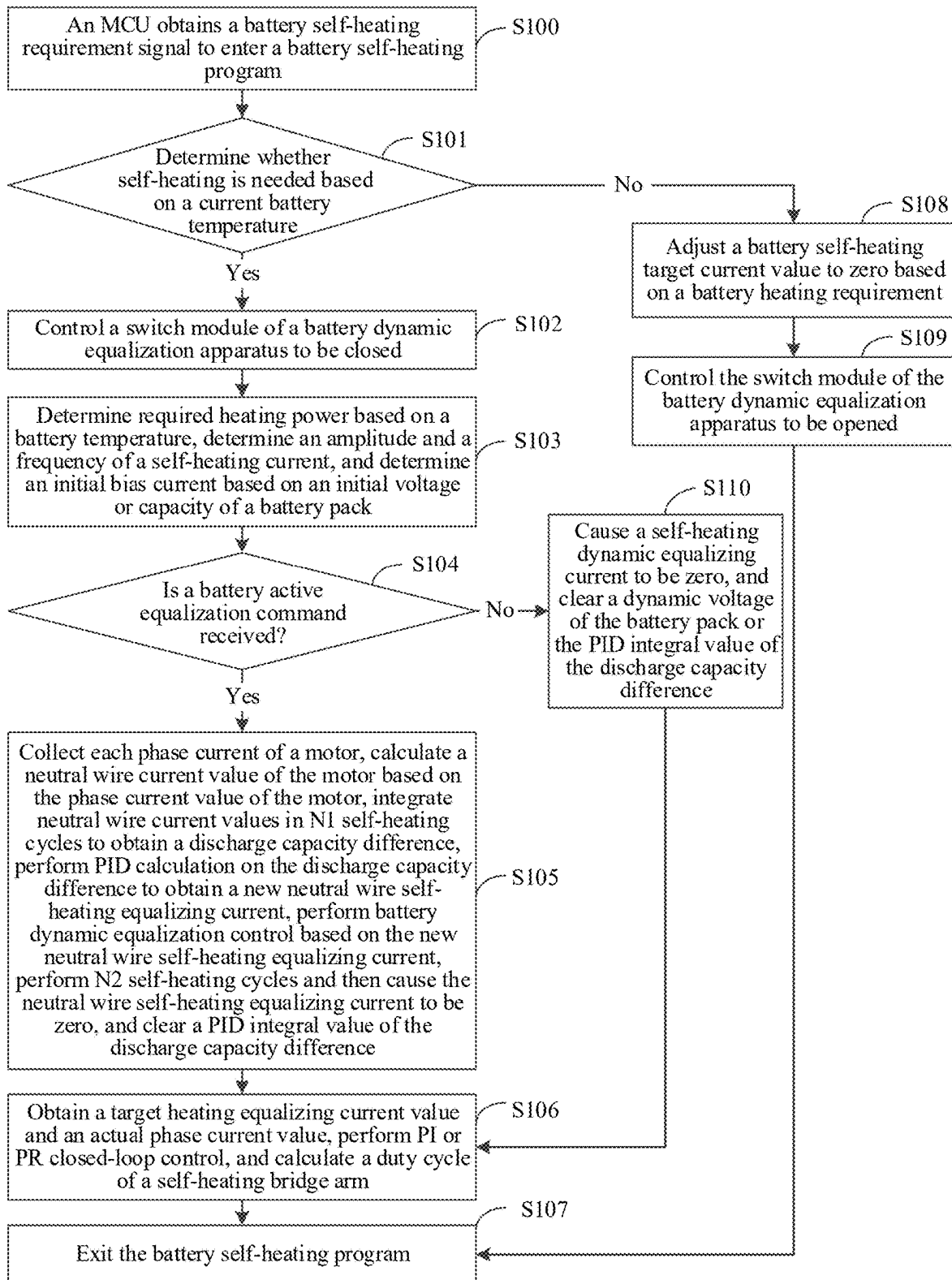


FIG. 3

Control a motor controller in a first preset state, to alternately charge and discharge a first battery pack and a second battery pack to implement self-heating of the first battery pack and the second battery pack, and cause an absolute value of a difference between a capacitance of the first battery pack and a capacitance of the second battery pack to be lower than a preset threshold

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FIG. 4



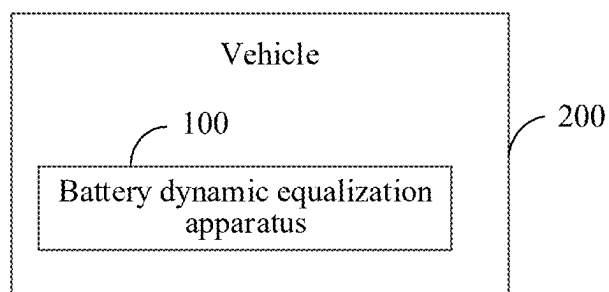


FIG. 6

**BATTERY DYNAMIC-EQUALIZATION
APPARATUS AND CONTROL METHOD
THEREFOR, AND VEHICLE**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

[0001] This application is a continuation application of International Patent Application No. PCT/CN2023/129009, filed on Nov. 1, 2023, which is based on and claims priority to and benefits of Chinese Patent Application No. 202211543573.X, filed on Nov. 30, 2022. The entire content of all of the above-referenced applications is incorporated herein by reference.

FIELD

[0002] The present disclosure relates to the field of battery protection technologies, and in particular, to a battery dynamic equalization apparatus, a control method for a battery dynamic equalization apparatus, and a vehicle.

BACKGROUND

[0003] With development and rapid popularization of electric vehicles, power batteries of the electric vehicles have become increasingly important. Currently, when a power battery of a vehicle is self-heating during driving of the vehicle, a voltage difference between batteries may gradually increase. This may lead to non-equalization between the batteries, and may reduce a battery capacity and shorten battery life.

SUMMARY

[0004] The present disclosure solves one of the technical problems in the related art at least to some extent. Therefore, a first aspect of the present disclosure is to provide a battery dynamic equalization apparatus, to implement dynamic equalization of electric quantity between a first battery pack and a second battery pack during self-heating of batteries when a vehicle is under a driving condition, thereby improving performance of the batteries and prolonging battery life.

[0005] A second aspect of the present disclosure is to provide a control method for a battery dynamic equalization apparatus.

[0006] A third aspect of the present disclosure is to provide a vehicle.

[0007] An embodiment of the first aspect of the present disclosure provides a battery dynamic equalization apparatus, including: a power battery, the power battery includes a first battery pack and a second battery pack that are connected in series, and a first node is provided between the first battery pack and the second battery pack; a motor controller, a first end of the motor controller is connected to a positive electrode of the first battery pack, and a second end of the motor controller is connected to a negative electrode of the second battery pack; a motor, where a first end of the motor is connected to the motor controller, and a second end of the motor is connected to the first node; and a controller, where the controller is connected to the motor controller, and the controller is configured to control, in a first preset state, a bridge arm of the motor controller to drive the motor, alternately charge and discharge the first battery pack and the second battery pack to implement self-heating of the first battery pack and the second battery pack, and to configure an absolute value of a difference between a capacitance of

the first battery pack and a capacitance of the second battery pack to be lower than a preset threshold.

[0008] In the battery dynamic equalization apparatus according to the embodiment of the present disclosure, the power battery includes the first battery pack and the second battery pack that are connected in series, and the first node is provided between the first battery pack and the second battery pack. The first end of the motor controller is connected to the positive electrode of the first battery pack, and the second end of the motor controller is connected to the negative electrode of the second battery pack. The first end of the motor is connected to the motor controller, and the second end of the motor is connected to the first node. The controller is connected to the motor controller. The controller is configured to control, in the first preset state, the bridge arm of the motor controller to drive the motor, alternately charge and discharge the first battery pack and the second battery pack to implement self-heating of the first battery pack and the second battery pack, and cause the absolute value of the difference between the capacitance of the first battery pack and the capacitance of the second battery pack to be lower than the preset threshold. Therefore, the equalization apparatus can implement dynamic equalization of electric quantity between the first battery pack and the second battery pack during self-heating of batteries when a vehicle is under a driving condition, thereby improving performance of the batteries and prolonging battery life.

[0009] In addition, the battery dynamic equalization apparatus according to the foregoing embodiment of the present disclosure may further have the following additional technical features.

[0010] According to an embodiment of the present disclosure, the motor controller includes at least two first bridge arms. A first end of each first bridge arm is connected to the positive electrode of the first battery pack, and a second end of each first bridge arm is connected to the negative electrode of the second battery pack. The motor includes at least two first inductors. A first end of each first inductor is connected to a corresponding first bridge arm, and a second end of each first inductor is connected to the first node. A neutral wire is led out from the motor, and the neutral wire is connected to the first node.

[0011] According to an embodiment of the present disclosure, the controller is configured to collect a battery equalization parameter, a temperature requirement parameter, and a driving requirement parameter of the power battery; obtain a target driving current based on the driving requirement parameter, obtain a target self-heating current based on the temperature requirement parameter, and obtain a target equalizing current based on the battery equalization parameter; and adjust, based on the target driving current, the target self-heating current, the target equalizing current, and an actual phase current, a first control signal for controlling the motor controller, to drive the motor, and self-heat and equalize the first battery pack and the second battery pack.

[0012] According to an embodiment of the present disclosure, the controller is configured to obtain a torque output of the motor, a rotational speed of the motor, a voltage of the power battery, and an electrical angle of a rotor of the motor; and obtain the target driving current based on the torque output of the motor, the rotational speed of the motor, and the voltage of the power battery, and obtain a differential mode voltage of each bridge arm based on the target driving

current, the actual phase current, and the electrical angle of the rotor of the motor, when a driving instruction is received.

[0013] According to an embodiment of the present disclosure, the target driving current is a synthesis of a d-axis target current and a q-axis target current, the actual phase current is a synthesis of a d-axis actual current and a q-axis actual current, and the controller is configured to perform closed-loop control on the d-axis target current, the d-axis actual current, the q-axis target current, and the q-axis actual current, to obtain a d-axis target voltage and a q-axis target voltage; and obtain the differential mode voltage of each bridge arm based on the d-axis target voltage, the q-axis target voltage, and the electrical angle of the rotor of the motor.

[0014] According to an embodiment of the present disclosure, the controller is further configured to obtain a target heating equalizing current based on the target self-heating current and the target equalizing current; obtain an actual phase current of the motor; obtain a common mode voltage based on the target heating equalizing current and the actual phase current; and adjust the first control signal of the motor controller based on the differential mode voltage of each bridge arm and the common mode voltage.

[0015] According to an embodiment of the present disclosure, the controller is configured to collect the battery equalization parameter and the temperature requirement parameter of the power battery; and obtain the target equalizing current based on the battery equalization parameter, and obtain the target self-heating current based on the temperature requirement parameter.

[0016] According to an embodiment of the present disclosure, the controller is configured to perform integral calculation on the battery equalization parameter to obtain the target equalizing current, where the battery equalization parameter includes: one of a discharge capacity difference, a voltage difference, and an instantaneous power difference, where the discharge capacity difference, the voltage difference, and the instantaneous power difference denote a discharge capacity difference, a voltage difference, and an instantaneous power difference between the first battery pack and the second battery pack, respectively.

[0017] According to an embodiment of the present disclosure, the battery equalization parameter is the discharge capacity difference; and the controller is configured to collect each phase current of the motor; calculate a neutral wire current of the motor based on each phase current of the motor; and integrate the neutral wire current to obtain the discharge capacity difference.

[0018] According to an embodiment of the present disclosure, the battery equalization parameter is the discharge capacity difference; and the controller is configured to collect a bus positive current and a bus negative current of the power battery; perform ampere-hour integration on the bus positive current of the power battery, to obtain a discharge capacity of the first battery pack, and perform ampere-time integration on the bus negative current of the power battery, to obtain a discharge capacity of the second battery pack; and obtain the discharge capacity difference based on the discharge capacity of the first battery pack and the discharge capacity of the second battery pack.

[0019] According to an embodiment of the present disclosure, the battery equalization parameter is the discharge capacity difference; and the controller is configured to collect a bus positive current of the motor controller and a

bus negative current of the motor controller; perform ampere-hour integration on the bus positive current, to obtain the discharge capacity of the first battery pack, and perform ampere-time integration on the bus negative current, to obtain the discharge capacity of the second battery pack; and obtain the discharge capacity difference based on the discharge capacity of the first battery pack and the discharge capacity of the second battery pack.

[0020] According to an embodiment of the present disclosure, the battery equalization parameter is the voltage difference; and the controller is configured to collect a bus voltage of the motor controller and a voltage of the first battery pack; calculate a voltage of the second battery pack based on the bus voltage and the voltage of the first battery pack; and determine the voltage difference based on the voltage of the first battery pack and the voltage of the second battery pack.

[0021] According to an embodiment of the present disclosure, the battery equalization parameter is the instantaneous power difference; and the controller is configured to collect the bus positive current and the battery bus negative current of the power battery, the voltage of the first battery pack, and the voltage of the second battery pack; determine instantaneous power of the first battery pack based on the battery bus positive current and the voltage of the first battery pack, and determine instantaneous power of the second battery pack based on the battery bus negative current and the voltage of the second battery pack; and determine the instantaneous power difference based on the instantaneous power of the first battery pack and the instantaneous power of the second battery pack.

[0022] According to an embodiment of the present disclosure, the controller is configured to obtain required heating power based on the temperature requirement parameter; determine an amplitude and a frequency of a self-heating current of the neutral wire based on the required heating power; and determine the target self-heating current based on the amplitude and the frequency.

[0023] According to an embodiment of the present disclosure, the controller is further configured to obtain the target heating equalizing current by the following formula:

$$In^* = indc + ipk \cdot \sin(2 \cdot \pi \cdot f \cdot t)$$

[0024] where In^* denotes the target heating equalizing current, $indc$ denotes the target equalizing current, ipk denotes an amplitude of the target self-heating current, f denotes a frequency of the target self-heating current, and t denotes a current time.

[0025] According to an embodiment of the present disclosure, the controller is configured to control, when it is detected that the absolute value of the difference between the capacitance of the first battery pack and the capacitance of the second battery pack is not lower than the preset threshold in a second preset state, the bridge arm of the motor controller to equalize the first battery pack and the second battery pack.

[0026] To achieve the foregoing objectives, an embodiment of the second aspect of the present disclosure provides a control method for a battery dynamic equalization apparatus. The apparatus includes: a power battery, a motor controller, a motor, and a controller, where the power battery

includes a first battery pack and a second battery pack that are connected in series, and a first node is provided between the first battery pack and the second battery pack. A first end of the motor controller is connected to a positive electrode of the first battery pack, and a second end of the motor controller is connected to a negative electrode of the second battery pack. A first end of the motor is connected to the motor controller, and a second end of the motor is connected to the first node. The controller is connected to the motor controller. The method is applied to the controller and includes: in a first preset state, the motor controller is controlled to drive the motor, alternately charge and discharge the first battery pack and the second battery pack to implement self-heating of the first battery pack and the second battery pack, and cause an absolute value of a difference between a capacitance of the first battery pack and a capacitance of the second battery pack to be lower than a preset threshold.

[0027] In the control method for a battery dynamic equalization apparatus according to the embodiment of the present disclosure, the apparatus includes: the power battery, the motor controller, the motor, and the controller, where the power battery includes the first battery pack and the second battery pack that are connected in series, and the first node is provided between the first battery pack and the second battery pack. The first end of the motor controller is connected to the positive electrode of the first battery pack, and the second end of the motor controller is connected to the negative electrode of the second battery pack. The first end of the motor is connected to the motor controller, and the second end of the motor is connected to the first node. The controller is connected to the motor controller. The method is applied to the controller and includes: in a first preset state, the motor controller is controlled to drive the motor, alternately charge and discharge the first battery pack and the second battery pack to implement self-heating of the first battery pack and the second battery pack, and cause the absolute value of the difference between the capacitance of the first battery pack and the capacitance of the second battery pack to be lower than the preset threshold. Therefore, the method can implement dynamic equalization of electric quantity between the first battery pack and the second battery pack during self-heating of batteries when a vehicle is under a driving condition, thereby improving performance of the batteries and prolonging battery life.

[0028] To achieve the foregoing objectives, an embodiment of the third aspect of the present disclosure provides a vehicle, including the above battery dynamic equalization apparatus.

[0029] In the vehicle according to the embodiment of the present disclosure, through the above battery dynamic equalization apparatus, dynamic equalization of electric quantity between the first battery pack and the second battery pack during self-heating of batteries when the vehicle is under a driving condition can be implemented, thereby improving performance of the batteries and prolonging battery life.

[0030] Some of additional aspects and advantages of the present disclosure are set forth in the following description, and some become apparent from the following description, or are learned by practice of the present disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] FIG. 1 is a schematic block diagram of a battery dynamic equalization apparatus according to an embodiment of the present disclosure;

[0032] FIG. 2 is a schematic circuit diagram of a battery dynamic equalization apparatus according to an embodiment of the present disclosure;

[0033] FIG. 3 is a schematic diagram of implementing battery dynamic equalization under a vehicle driving condition and a self-heating mode according to an embodiment of the present disclosure;

[0034] FIG. 4 is a flowchart of a control method for a battery dynamic equalization apparatus according to an embodiment of the present disclosure;

[0035] FIG. 5 is a flowchart of a control method for a battery dynamic equalization apparatus according to an example of the present disclosure; and

[0036] FIG. 6 is a schematic block diagram of a vehicle according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0037] Embodiments of the present disclosure are described in detail below, and examples of the embodiments are shown in accompanying drawings. The same or similar elements or the elements having same or similar functions are denoted by the same or similar reference signs throughout the description. The embodiments described below with reference to the accompanying drawings are examples, and are to explain the present disclosure and cannot be construed as a limitation to the present disclosure.

[0038] During self-heating operation of a vehicle battery, each battery pack is constantly charging and discharging at a high frequency. In fact, there are differences in state, impedance, temperature characteristics, and the like of each battery unit. Once a voltage difference between battery packs occurs, if no equalization regulation is performed, non-equalization between the battery packs is accelerated. As a self-heating time increases, the voltage difference between battery packs also gradually increases, battery capacities become increasingly un-equalized (for example, there are two battery packs E1 and E2, one of the battery packs is fully charged, and the other is fed), and a mile range sharply decreases. This may shorten battery life in a long time. Therefore, to solve non-equalization of voltage between battery packs during self-heating of batteries, the present disclosure provides a battery dynamic equalization apparatus. When a vehicle is under a driving condition, before a self-heating function of the batteries is enabled, whether an initial electric quantity of each battery pack is equalized and whether the battery packs are equalized during self-heating operation of the batteries are determined, so that dynamic equalization of electric quantity between the battery packs can be implemented when the battery packs are non-equalized in electric quantity during self-heating of the batteries. This improves performance of the batteries and prolongs battery life.

[0039] A battery dynamic equalization apparatus, a control method for a battery dynamic equalization apparatus, and a vehicle according to embodiments of the present disclosure are described below with reference to the accompanying drawings.

[0040] FIG. 1 is a schematic block diagram of a battery dynamic equalization apparatus according to an embodiment of the present disclosure.

[0041] As shown in FIG. 1, the battery dynamic equalization apparatus may include: a power battery 110, a motor controller 120, a motor 130, and a controller 140.

[0042] The power battery 110 includes a first battery pack E1 and a second battery pack E2 that are connected in series, and a first node J1 is provided between the first battery pack E1 and the second battery pack E2. A first end of the motor controller 120 is connected to a positive electrode of the first battery pack E1, and a second end of the motor controller 120 is connected to a negative electrode of the second battery pack E2. A first end of the motor 130 is connected to the motor controller 120, and a second end of the motor 130 is connected to the first node J1. The controller 140 is connected to the motor controller, and the controller 140 is configured to control, in a first preset state, a bridge arm of the motor controller 120 to drive the motor 130, alternately charge and discharge the first battery pack E1 and the second battery pack E2 to implement self-heating of the first battery pack E1 and the second battery pack E2, and cause an absolute value of a difference between a capacitance of the first battery pack E1 and a capacitance of the second battery pack E2 to be lower than a preset threshold. The first preset state is a state of self-heating and equalization of the power battery when a vehicle is under a driving condition. The preset threshold may be determined based on an actual case.

[0043] Further, as shown in FIG. 2, according to an embodiment of the present disclosure, the motor controller 120 includes at least two phases of first bridge arms 121. A first end of each phase of first bridge arm 121 is connected to the positive electrode of the first battery pack E1, and a second end of each phase of first bridge arm 121 is connected to the negative electrode of the second battery pack E2. The motor 130 includes at least two phases of first inductors 131. A first end of each phase of first inductor 131 is connected to a corresponding first bridge arm 121, and a second end of each phase of first inductor 131 is connected to the first node J1. A neutral wire is led out from the motor 130, and the neutral wire is connected to the first node J1.

[0044] In an embodiment, the motor 130 may include a permanent magnet synchronous motor, an asynchronous motor, or other types of motor. The motor 130 may be three-phase, five-phase, six-phase, nine-phase, or the like. The motor 130 may include a form of multiple windings. A motor coil includes x windings, where x is greater than or equal to 1 and x is an integer. Each winding includes at least two phases of windings, and each phase of winding corresponds to a group of bridge arms for control. A winding shown in FIG. 2 and a motor being a three-phase motor are taken as an example for illustration. Two battery packs (a first battery pack E1 and a second battery pack E2) of the power battery 110 are first connected in series, and each battery pack is formed by connecting multiple battery units (or cells) in series. The two battery packs include the same battery units. The battery units of the entire power battery 110 are even-numbered. A positive bus, a negative bus, and a battery pack series-connection midpoint lead-out wire are led out from the power battery 110. The positive bus and the negative bus of the power battery 110 are connected to a positive bus and a negative bus of the motor controller 120 by switches K2 and K1 respectively. The motor controller 120 includes at least one phase of first bridge arm 121. For

example, in FIGS. 2, A, B, and C are first bridge arms 121, which are connected to a three-phase coil (first inductor 131) of the motor 130, and a lead-out neutral wire (neutral wire) of the motor 130 is connected to a switch K6. The switch K6 is connected to a series-connection midpoint J1 lead-out wire of the power battery 110, and a bus capacitor C1 is connected in parallel to positive and negative buses of the motor controller 120. A positive series connection switch K5 of a DC charging and discharging port is connected to a bus positive electrode of the motor controller 120, and a positive series connection switch K4 of the DC charging and discharging port is connected to the lead-out neutral wire of the motor. A negative series connection switch K3 of the DC charging and discharging port is connected to a bus negative electrode of the motor controller 120. A positive electrode of a capacitor C2 is connected to a positive electrode of the DC charging and discharging port, and a negative electrode of the capacitor C2 is connected to the bus negative electrode of the motor controller 120. In a shutdown state, that is, when the vehicle is neither charged nor driving, and a system is powered off, the switches K1, K2, K3, K4, K5, and K6 are all open. When the vehicle is in the first preset state, that is, the vehicle is under any working condition, and when a self-heating requirement signal is received, the switch K6 is controlled to be closed, and the power battery 110 may perform self-heating. During self-heating, each battery pack is constantly charged and discharged, and the controller 140 (not shown in the figure) may adjust and control a duty cycle of each phase of bridge arm of the motor controller 120, to equalize the first battery pack E1 and the second battery pack E2. For example, when the vehicle is under a driving condition, and when the first battery pack E1 has a high capacitance while the second battery pack E2 has a low capacitance, the duty cycle for connection of an upper bridge arm can be controlled to be higher, so that electricity of the first battery pack E1 may be transferred to the first inductor in the motor 130 for storage, and is discharged to the second battery pack E2 through the lead-out neutral wire of the motor. That is, by alternate charging and discharging of the first battery pack E1 and the second battery pack E2, the absolute value of the difference between the capacitance of the first battery pack E1 and the capacitance of the second battery pack E2 is reduced to be lower than the preset threshold, so that dynamic equalization of electric quantity between the first battery pack and the second battery pack during self-heating of batteries can be implemented, thereby improving performance of the batteries and prolonging battery life. The duty cycle refers to a duty cycle for connection of the upper bridge arm, and duty cycles of the upper bridge arm and a lower bridge arm are complementary. For example, if the upper bridge arm is connected for 70% of total duration, the lower bridge arm is connected for 30% thereof, so that a battery pack having a high capacitance is more discharged, while a battery pack having a low capacitance is less discharged, and a certain dead time may be retained based on characteristics of a power device.

[0045] The battery dynamic equalization apparatus according to the present disclosure is described in detail below.

[0046] According to an embodiment of the present disclosure, the controller 140 is configured to collect a battery equalization parameter, a temperature requirement parameter, and a driving requirement parameter of the power battery 110; obtain a target driving current based on the

driving requirement parameter, obtain a target self-heating current based on the temperature requirement parameter, and obtain a target equalizing current based on the battery equalization parameter; and adjust, based on the target driving current, the target self-heating current, the target equalizing current, and an actual phase current, a first control signal for controlling the motor controller, to drive the motor, and self-heat and equalize the first battery pack E1 and the second battery pack E2.

[0047] In an embodiment, the controller 140 is configured to collect the battery equalization parameter of the power battery 110 when the battery packs need to be equalized, the temperature requirement parameter of the power battery during self-heating of the power battery 110, and the driving requirement parameter of the vehicle when the vehicle is driving. After the battery equalization parameter, the temperature requirement parameter, and the driving requirement parameter are obtained, the target driving current may be obtained based on the driving requirement parameter. For example, the target driving current may be obtained based on a driving torque of the motor 130, a rotational speed of the motor, a bus voltage, and the like. The target self-heating current is obtained based on the temperature requirement parameter. For example, a table look-up method may be used. In a table, the temperature requirement parameter and the target self-heating current are in a one-to-one correspondence, and the target self-heating current may be determined based on the temperature requirement parameter. A larger temperature requirement parameter indicates a larger target self-heating current. A smaller temperature requirement parameter indicates a smaller target self-heating current. The target equalizing current is obtained based on the battery equalization parameter. After the target driving current, the target self-heating current, and the target equalizing current are obtained, the first control signal for controlling the motor controller may be adjusted based on the target driving current, the target self-heating current, the target equalizing current, and a current actual phase current. For example, when the vehicle is under a driving condition, and the power battery 110 is self-heating, the first battery pack E1 has a higher capacitance while the second battery pack E2 has a lower capacitance, and the first battery pack E1 may be more discharged and the second battery pack E2 may be less discharged through the first control signal for controlling the motor controller. Therefore, when the vehicle is under the driving condition and the power battery is self-heating, the capacitances of the first battery pack E1 and the second battery pack E2 tend to be equalized.

[0048] According to an embodiment of the present disclosure, the controller 140 is configured to obtain a torque output of the motor 130, a rotational speed of the motor 130, a voltage of the power battery 110, and an electrical angle of a rotor of the motor 130; and obtain the target driving current based on the torque output of the motor 130, the rotational speed of the motor 130, and the voltage of the power battery 110, and obtain a differential mode voltage of each phase of bridge arm based on the target driving current, the actual phase current, and the electrical angle of the rotor of the motor 130, when a driving instruction is received.

[0049] Further, according to an embodiment of the present disclosure, the target driving current is a synthesis of a d-axis target current and a q-axis target current, the actual phase current is a synthesis of a d-axis actual current and a q-axis actual current, and the controller 140 is configured to

perform closed-loop control on the d-axis target current, the d-axis actual current, the q-axis target current, and the q-axis actual current, to obtain a d-axis target voltage and a q-axis target voltage; and obtain the differential mode voltage of each phase of bridge arm based on the d-axis target voltage, the q-axis target voltage, and the electrical angle of the rotor of the motor 130.

[0050] In an embodiment, during control of the permanent magnet synchronous motor, to obtain control characteristics similar to those of a direct current motor, a coordinate system is established on a rotor of the motor. The coordinate system synchronously rotates with the rotor. A direction of a magnetic field of the rotor is a d-axis, and a direction perpendicular to the magnetic field of the rotor is a q-axis. A mathematical model of the motor may be converted under the coordinate system, to implement decoupling of the d-axis and the q-axis, so as to obtain good control characteristics. That is, coordinate transformation may be performed on each phase current of the motor 130, to obtain a d-axis current and a q-axis current of the motor. The d-axis current and the q-axis current are used to drive the vehicle. As shown in FIG. 3, the motor has a total of p phase currents, i_1, i_2 to i_p . After coordinate transformation is performed on each phase current, a d-axis current i_d and a q-axis current i_q of the motor are obtained. The actual phase current is a synthesis of an actual d-axis current i_d and an actual q-axis current i_q . A torque output T_e^* of the motor, a rotational speed W of the motor, a voltage U_{dc} of the power battery 110, and an electrical angle θ of a rotor of the motor are obtained. When a driving instruction is received, and the vehicle is under a driving condition, the target driving current may be obtained based on the torque output T_e^* of the motor, the rotational speed W of the motor, and the voltage U_{dc} of the power battery 110. The target driving current is a synthesis of a d-axis target current and a q-axis target current. A d-axis target voltage U_d may be obtained by performing PID closed-loop control based on a d-axis target current i_d^* and a d-axis actual current i_d , and a q-axis target voltage U_q may be obtained by performing PID closed-loop control based on a q-axis target current i_q^* and a q-axis actual current i_q . After the d-axis target voltage U_d and the q-axis target voltage U_q are obtained, a differential mode voltage $U_1, U_2, \dots, U_p$ of each phase of bridge arm is obtained after inverse coordinate transformation is performed based on the target voltage U_d , the target voltage U_q , and the electrical angle θ of the rotor of the motor.

[0051] According to an embodiment of the present disclosure, the controller 140 is further configured to obtain a target heating equalizing current based on the target self-heating current and the target equalizing current; obtain an actual phase current of the motor; obtain a common mode voltage based on the target heating equalizing current and the actual phase current; and adjust the first control signal of the motor controller based on the differential mode voltage of each phase of bridge arm and the common mode voltage.

[0052] In an embodiment, as shown in FIG. 3, after the vehicle receives a self-heating power instruction and a battery equalization instruction, a target heating equalizing current i_n^* is obtained based on the target self-heating current and a target equalizing current i_{nc} through an instruction solving module 2. For example, in a p-phase (p+1)-wire motor, a current of each phase of the motor is collected, to obtain p phase currents ($i_1, i_2, \dots, i_p$). A direction in which the current flows into the motor is a

positive direction, and $i_1+i_2+\dots+i_p+i_n=0$ is satisfied, where i_n is a neutral wire current of the motor. A magnitude of each phase current may be collected by a current sensor, and after each phase current of the motor is obtained, a neutral wire current i_n of the motor may be calculated through $i_n=-i_1-i_2-\dots-i_p$. The common mode voltage is obtained based on the target heating equalizing current i_n^* and the actual phase current. That is, after the actual neutral wire current i_n is obtained, a common mode voltage U_n controlled by p phases of bridge arms is obtained by PR closed-loop control based on the target heating equalizing current i_n^* and the actual neutral wire current i_n . A PWM duty cycle of each phase in the p phases of bridge arms is adjusted based on the common mode voltage U_n and a differential mode voltage ($U_1, U_2, \dots, U_p$) of each phase of bridge arm of the motor, to adjust the first control signal of the motor controller, so that the vehicle implements self-heating of the power battery and equalization of electric quantity between battery packs under a driving condition.

[0053] According to an embodiment of the present disclosure, the controller **140** is configured to collect the battery equalization parameter and the temperature requirement parameter of the power battery **110**; and obtain the target equalizing current based on the battery equalization parameter, and obtain the target self-heating current based on the temperature requirement parameter.

[0054] Further, according to an embodiment of the present disclosure, the controller **140** is configured to perform integral calculation on the battery equalization parameter to obtain the target equalizing current, where the battery equalization parameter includes: one of a discharge capacity difference, a voltage difference, and an instantaneous power difference, where the discharge capacity difference, the voltage difference, and the instantaneous power difference denote a discharge capacity difference, a voltage difference, and an instantaneous power difference between the first battery pack **E1** and the second battery pack **E2**, respectively.

[0055] In an embodiment, the controller **140** collects the battery equalization parameter of the power battery **110** and the temperature requirement parameter of the power battery **110**. The battery equalization parameter of the power battery **110** may include a discharge capacity difference ΔQ . Integral calculation is performed on the discharge capacity difference ΔQ to obtain a target equalizing current. After the target equalizing current is obtained, a target self-heating current may be obtained based on the temperature requirement parameter when the power battery **110** is self-heating. For example, if there is a larger current temperature requirement parameter, there is a higher target self-heating current, or if there is a smaller current temperature requirement parameter, there is a lower target self-heating current.

[0056] The battery equalization parameter of the power battery **110** may further include a voltage difference ΔU . A target equalizing current is obtained by integrating the voltage difference ΔU between the first battery pack **E1** and the second battery pack **E2**, and a target self-heating current is obtained based on the temperature requirement parameter when the power battery **110** is self-heating. For example, if there is a larger current temperature requirement parameter, there is a higher target self-heating current, or if there is a smaller current temperature requirement parameter, there is a lower target self-heating current.

[0057] The battery equalization parameter of the power battery **110** may further include an instantaneous power difference ΔP . A target equalizing current is obtained by integrating the instantaneous power difference ΔP between the first battery pack **E1** and the second battery pack **E2**, and a target self-heating current is obtained based on the temperature requirement parameter when the power battery **110** is self-heating. For example, if there is a larger current temperature requirement parameter, there is a higher target self-heating current, or if there is a smaller current temperature requirement parameter, there is a lower target self-heating current.

[0058] According to an embodiment of the present disclosure, the battery equalization parameter is the discharge capacity difference; and the controller **140** is configured to collect each phase current of the motor **130**; calculate a neutral wire current of the motor **130** based on each phase current of the motor **130**; and integrate the neutral wire current to obtain the discharge capacity difference.

[0059] In an embodiment, the motor **130** is an M-phase N-wire motor. For example, in a six-phase seven-wire motor, a current of each phase of the motor **130** is collected, to obtain six phase currents (i_a, i_b, i_c, i_u, i_v , and i_w) are obtained. Taking a direction in which the current flows into the motor **130** as a positive direction, $i_a+i_b+i_c+i_u+i_v+i_w+i_n=0$ is satisfied, where i_n is a neutral wire current of the motor **130**. Each phase current of the motor **130** is collected by the controller **140**. For example, magnitudes of phase currents i_a, i_b, i_c, i_u, i_v , and i_w may be collected by a current sensor, and after each phase current of the motor **130** is obtained, a neutral wire current i_n of the motor **130** may be calculated through $i_n=-i_a-i_b-i_c-i_u-i_v-i_w$. After the neutral wire current i_n of the motor **130** is obtained, the neutral wire current i_n may be integrated, to obtain the discharge capacity difference ΔQ between the first battery pack **E1** and the second battery pack **E2**. A phase shift control technology (troughs and peaks of PWM count values of each phase of bridge arm are sampled, and phase shift control can implement multiple level sampling of a carrier cycle) is used for motor control. An MCU collects each phase current of the motor through a zero-crossing point and a cycle point of the carrier cycle of each phase of bridge arm, which is used to calculate the neutral wire current of the motor, improve sampling times in a unit cycle of the neutral wire current of the motor, and calculate a discharge difference between the two battery packs more accurately. Through the phase shift control technology, a current ripple of the neutral wire of the motor can be reduced, and certain electro magnetic compatibility (EMC) interference and noise vibration harshness (NVH) noise can be reduced. An update cycle of the discharge capacity difference ΔQ is based on a carrier cycle of each bridge arm. In addition, each phase current of the motor and the neutral wire current of the motor are sampled, and the integration of the neutral wire current of the motor during a self-heating cycle is calculated and is updated once.

[0060] According to an embodiment of the present disclosure, the battery equalization parameter is the discharge capacity difference; and the controller **140** is configured to collect a bus positive current and a bus negative current of the power battery **110**; perform ampere-hour integration on the bus positive current of the power battery **110**, to obtain a discharge capacity of the first battery pack **E1**, and perform ampere-time integration on the bus negative current of the power battery **110**, to obtain a discharge capacity of the

second battery pack E2; and obtain the discharge capacity difference based on the discharge capacity of the first battery pack E1 and the discharge capacity of the second battery pack E2.

[0061] In an embodiment, during self-heating of the power battery 110, the controller 140 collects the bus positive current and the bus negative current of the power battery 110, performs ampere-hour integration on the bus positive current and the bus negative current of the power battery 110, that is, performs ampere-hour integration on the bus negative current of the power battery 110 to calculate a cumulative discharge capacity Q2 and performs ampere-hour integration on the bus positive current of the power battery 110 to calculate a cumulative discharge capacity Q1, and may obtain the discharge capacity difference ΔQ based on the discharge capacity Q1 of the first battery pack E1 and the discharge capacity Q2 of the second battery pack E2 after obtaining the discharge capacities Q1 and Q2. The bus positive current and the bus negative current of the power battery terminal may be collected by a battery management system (BMS), and current Hall of a bus positive electrode and a bus negative electrode of a battery pack and a voltage sampling circuit of each battery unit are directly integrated in the BMS without additional devices.

[0062] According to an embodiment of the present disclosure, the battery equalization parameter is the discharge capacity difference; and the controller 140 is configured to collect a bus positive current of the motor controller 120 and a bus negative current of the motor controller 120; perform ampere-hour integration on the bus positive current of the motor controller 120, to obtain the discharge capacity of the first battery pack E1, and perform ampere-time integration on the bus negative current of the motor controller 120, to obtain the discharge capacity of the second battery pack E2; and obtain the discharge capacity difference based on the discharge capacity of the first battery pack E1 and the discharge capacity of the second battery pack E2.

[0063] In an embodiment, during self-heating of the power battery 110, the controller 140 collects the bus positive current and the bus negative current of the motor controller 120, performs ampere-hour integration on the bus positive current and the bus negative current of the motor controller 120, that is, performs ampere-hour integration on the bus positive current of the motor controller 120 to calculate a cumulative discharge capacity Q1 and performs ampere-hour integration on the bus negative current of the motor controller 120 to calculate a cumulative discharge capacity Q2, and may obtain the discharge capacity difference ΔQ based on the discharge capacity Q1 of the first battery pack E1 and the discharge capacity Q2 of the second battery pack E2 after obtaining the discharge capacities Q1 and Q2. The bus positive current of the of the motor controller 120 and the bus negative current of the motor controller 120 may be collected by the MCU, which can avoid a switching time of a power device and interference, to collect the currents more accurately.

[0064] According to an embodiment of the present disclosure, the battery equalization parameter is the voltage difference, and the controller 140 is configured to collect a bus voltage of the motor controller 120 and a voltage of the second battery pack E2; calculate a voltage of the first battery pack E1 based on the bus voltage and the voltage of the second battery pack E2; and determine the voltage

difference based on the voltage of the first battery pack E1 and the voltage of the second battery pack E2.

[0065] In an embodiment, the controller 140 may collect the bus voltage of the motor controller 120 and a neutral wire voltage to earth of the motor (the voltage of the second battery pack E2), where the bus voltage of the motor controller 120 is a sum of the voltage of the first battery pack E1 and the voltage of the second battery pack E2. After the bus voltage of the motor controller 120 and the voltage of the second battery pack E2 are obtained, the voltage of the first battery pack E1 may be calculated by subtracting the voltage of the second battery pack E2 from the bus voltage of the motor controller 120. After the voltage of the first battery pack E1 and the voltage of the second battery pack E2 are obtained, the voltage difference ΔU may be determined based on the voltage of the first battery pack E1 and the voltage of the second battery pack E2.

[0066] According to an embodiment of the present disclosure, the battery equalization parameter is the instantaneous power difference; and the controller 140 is configured to collect the bus positive current and the battery bus negative current of the power battery 110, the voltage of the first battery pack E1, and the voltage of the second battery pack E2; determine instantaneous power of the first battery pack E1 based on the battery bus positive current and the voltage of the first battery pack E1, and determine instantaneous power of the second battery pack based on the battery bus negative current and the voltage of the second battery pack; and determine the instantaneous power difference based on the instantaneous power of the first battery pack and the instantaneous power of the second battery pack.

[0067] In an embodiment, the controller 140 collects the bus positive current of the power battery 110, the bus negative current of the power battery, the voltage of the first battery pack E1, and the voltage of the second battery pack E2, may determine instantaneous power p1 of the first battery pack E1 based on a product of the voltage of the first battery pack E1 and the bus positive current of the power battery 110, may determine instantaneous power p2 of the second battery pack E2 based on a product of the voltage of the second battery pack E2 and the bus negative current of the power battery, and may determine the instantaneous power difference Δp based on the instantaneous power p1 of the first battery pack E1 and the instantaneous power p2 of the second battery pack E2 after calculating the instantaneous power of each battery pack.

[0068] According to an embodiment of the present disclosure, the controller 140 is configured to obtain required heating power based on the temperature requirement parameter; determine an amplitude and a frequency of a self-heating current of the neutral wire based on the required heating power; and determine the target self-heating current based on the amplitude and the frequency.

[0069] Further, according to an embodiment of the present disclosure, the controller 140 is further configured to obtain the target heating equalizing current by the following formula:

$$In^* = indc + ipk \cdot \sin(2 \cdot \pi \cdot f \cdot t), \quad (1)$$

[0070] where In^* denotes the target heating equalizing current, $indc$ denotes the target equalizing current, ipk

denotes an amplitude of the target self-heating current, f denotes a frequency of the target self-heating current, and t denotes a current time.

[0071] In an embodiment, after a self-heating requirement signal of the power battery 110 is obtained, whether a self-heating function needs to be enabled may be determined based on a current temperature of the power battery 110. When the current temperature of the power battery 110 is low, to ensure the power supply capacity of the power battery 110, a self-heating program of the power battery may be entered, and required heating power may be determined based on a heating requirement for the power battery 110, that is, based on the temperature requirement parameter when the power battery 110 is self-heating. For example, if the current temperature of the power battery 110 is lower, the required heating power is higher. For example, if the current temperature of the power battery 110 is higher, the required heating power is lower. After the required heating power is obtained, the amplitude ipk and the frequency f of the self-heating current of the neutral wire may be determined based on an amplitude of the required heating power, and the target self-heating current may be determined based on the amplitude ipk and the frequency f after the amplitude ipk and the frequency f of the self-heating current of the neutral wire are obtained. The target heating equalizing current In^* may be determined based on the target equalizing current $indc$, the amplitude ipk and the frequency f of the target self-heating current, and the current time through the above formula (1).

[0072] It should be noted that, the temperature requirement parameter of the power battery 110, the amplitude ipk of the target self-heating current, and the required heating power corresponding to the frequency f of the target self-heating current may be calibrated on a bench first, and may generally be obtained by a table look-up method or a linear fitting method.

[0073] According to an embodiment of the present disclosure, the controller 140 is configured to control, when it is detected that the absolute value of the difference between the capacitance of the first battery pack E1 and the capacitance of the second battery pack E2 is not lower than the preset threshold in a second preset state, the bridge arm of the motor controller to equalize the first battery pack E1 and the second battery pack E2.

[0074] In an embodiment, when it is detected that the absolute value of the difference between the capacitance of the first battery pack E1 and the capacitance of the second battery pack E2 is not lower than the preset threshold, it means that the current difference between the capacitance of the first battery pack E1 and the capacitance of the second battery pack E2 is large, and the power battery 110 is not in an equalized state. That is, in the second preset state (the state in which the power battery 110 only performs electric quantity equalization), the bridge arm of the motor controller 120 may be controlled to equalize the first battery pack E1 and the second battery pack E2, so that the absolute value of the difference between the capacitance of the first battery pack E1 and the capacitance of the second battery pack E2 is smaller. For example, when the first battery pack E1 has a larger capacitance while the second battery pack E2 has a smaller capacitance, a discharging time of the first battery pack E1 may be longer and a charging time of the second

battery pack E2 may be longer, to implement the equalization of capacitance between the first battery pack E1 and the second battery pack E2.

[0075] To sum up, in the battery dynamic equalization apparatus according to the embodiment of the present disclosure, the power battery includes the first battery pack and the second battery pack that are connected in series, and the first node is provided between the first battery pack and the second battery pack. The first end of the motor controller is connected to the positive electrode of the first battery pack, and the second end of the motor controller is connected to the negative electrode of the second battery pack. The first end of the motor is connected to the motor controller, and the second end of the motor is connected to the first node. The controller is connected to the motor controller. The controller is configured to control, in the first preset state, the bridge arm of the motor controller to drive the motor, alternately charge and discharge the first battery pack and the second battery pack to implement self-heating of the first battery pack and the second battery pack, and cause the absolute value of the difference between the capacitance of the first battery pack and the capacitance of the second battery pack to be lower than the preset threshold. Therefore, the equalization apparatus can implement dynamic equalization of electric quantity between the first battery pack and the second battery pack during self-heating of batteries when a vehicle is under a driving condition, thereby improving performance of the batteries and prolonging battery life.

[0076] Corresponding to the foregoing embodiment, the present disclosure further provides a control method for a battery dynamic equalization apparatus.

[0077] As shown in FIG. 4, the control method for a battery dynamic equalization apparatus according to an embodiment of the present disclosure includes the following steps:

[0078] S1: In a first preset state, a motor controller is controlled, to alternately charge and discharge a first battery pack and a second battery pack to implement self-heating of the first battery pack and the second battery pack, and cause an absolute value of a difference between a capacitance of the first battery pack and a capacitance of the second battery pack to be lower than a preset threshold.

[0079] The control method according to the present disclosure is described with reference to FIG. 5 below.

[0080] As an example, the control method for a battery dynamic equalization apparatus according to the present disclosure may include the following steps:

[0081] S100: An MCU obtains a battery self-heating requirement signal to enter a battery self-heating program.

[0082] S101: Determine whether self-heating is needed based on a current battery temperature. If yes, execute step S102, or if not, execute step S108.

[0083] S102: Control a switch module of the battery dynamic equalization apparatus to be closed.

[0084] S103: Determine required heating power based on a battery temperature, determine an amplitude and a frequency of a self-heating current, and determine an initial bias current based on an initial voltage or capacity of a battery pack.

[0085] S104: Determine whether a battery active equalization command is received. If yes, execute step S105, or if not, execute step S110.

[0086] S105: Collect each phase current of a motor, calculate a neutral wire current value of the motor based on the

phase current value of the motor, integrate neutral wire current values in N1 self-heating cycles to obtain a discharge capacity difference, perform PID calculation on the discharge capacity difference to obtain a new neutral wire self-heating equalizing current, perform battery dynamic equalization control based on the new neutral wire self-heating equalizing current, perform N2 self-heating cycles and then cause the neutral wire self-heating equalizing current to be zero, and clear a PID integral value of the discharge capacity difference, where no PID calculation on the discharge capacity difference is performed during the N2 self-heating cycles, N1 and N2 are positive integers, and N2 is much greater than N1 (for example, $N2=100N1$).

[0087] S106: Obtain a target heating equalizing current value and an actual phase current value, perform PI or PR closed-loop control, and calculate a duty cycle of a self-heating bridge arm.

[0088] S107: Exit the battery self-heating program.

[0089] S108: Adjust a battery self-heating target current value to zero based on a battery heating requirement.

[0090] S109: Control the switch module of the battery dynamic equalization apparatus to be opened, and perform step S107.

[0091] S110: Cause a self-heating dynamic equalizing current to be zero, clear a dynamic voltage of the battery pack or the PID integral value of the discharge capacity difference, and perform step S106.

[0092] It should be noted that, for details not disclosed in the control method for a battery dynamic equalization apparatus according to the embodiment of the present disclosure, refer to the details disclosed in the battery dynamic equalization apparatus according to the embodiment of the present disclosure. Details are not described herein.

[0093] In the control method for a battery dynamic equalization apparatus according to the embodiment of the present disclosure, the apparatus includes: a power battery, a motor controller, a motor, and a controller, where the power battery includes a first battery pack and a second battery pack that are connected in series, and a first node is provided between the first battery pack and the second battery pack. A first end of the motor controller is connected to a positive electrode of the first battery pack, and a second end of the motor controller is connected to a negative electrode of the second battery pack. A first end of the motor is connected to the motor controller, and a second end of the motor is connected to the first node. The controller is connected to the motor controller. The method is applied to the controller and includes: in the first preset state, the motor controller is controlled to drive the motor, alternately charge and discharge the first battery pack and the second battery pack to implement self-heating of the first battery pack and the second battery pack, and cause the absolute value of the difference between the capacitance of the first battery pack and the capacitance of the second battery pack to be lower than the preset threshold. Therefore, the method can implement dynamic equalization of electric quantity between the first battery pack and the second battery pack during self-heating of batteries when a vehicle is under any condition, thereby improving performance of the batteries and prolonging battery life.

[0094] Corresponding to the foregoing embodiments, the present disclosure further provides a vehicle.

[0095] As shown in FIG. 6, a vehicle 200 according to an embodiment of the present disclosure may include the above battery dynamic equalization apparatus 100.

[0096] In the vehicle according to the embodiment of the present disclosure, through the above battery dynamic equalization apparatus included, dynamic equalization of electric quantity between the first battery pack and the second battery pack during self-heating of batteries when the vehicle is under a driving condition can be implemented, thereby improving performance of the batteries and prolonging battery life.

[0097] It should be noted that, the logic and/or steps shown in the flowcharts or described in any other manner herein, for example, a sequenced list that may be considered as executable instructions used for implementing logical functions, may be implemented in any computer-readable medium to be used by an instruction execution system, apparatus, or device (for example, a computer-based system, a system including a processor, or another system that can obtain an instruction from the instruction execution system, apparatus, or device and execute the instruction) or to be used by combining such instruction execution systems, apparatuses, or devices. In the context of this specification, a “computer-readable medium” may be any apparatus that can include, store, communicate, propagate, or transmit the program for use by the instruction execution system, apparatus, or device or in combination with the instruction execution system, apparatus, or device. More examples (a non-exhaustive list) of computer-readable media include: an electrical connection piece (electronic apparatus) having one or more wires, a portable computer disk cartridge (magnetic apparatus), a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or flash memory), an optical fiber apparatus, and a portable compact disc read-only memory (CD-ROM). In addition, the computer-readable medium can even be paper or other suitable media on which the program can be printed, because the program can be obtained electronically by, for example, optically scanning paper or other media, then editing, interpreting, or processing in other suitable ways if necessary, and then storing it in a computer memory.

[0098] It should be understood that, parts of the present disclosure can be implemented by using hardware, software, firmware, or a combination thereof. In the foregoing implementations, multiple steps or methods may be implemented by using software or firmware that are stored in a memory and are executed by a proper instruction execution system. For example, if they are implemented by hardware, as in another implementation, they can be implemented by any one of the following technologies well known in the art or a combination thereof: a discrete logic circuit including a logic gate circuit for implementing a logic function of a data signal, a dedicated integrated circuit including a proper combined logic gate circuit, a programmable gate array (PGA), a field programmable gate array (FPGA), and the like.

[0099] In the description of this specification, descriptions with reference to the terms such as “an embodiment”, “some embodiments”, “example”, “specific example” or “some examples” mean that features, structures, materials, or characteristics described with reference to the embodiment or example are included in at least one embodiment or example of the present disclosure. In this specification, examples of

descriptions of the foregoing terms do not necessarily refer to the same embodiment or example. In addition, the described features, structures, materials, or characteristics may be combined in a proper manner in any one or more of the embodiments or examples.

[0100] In addition, terms “first” and “second” are used merely for the purpose of description, and shall not be construed as indicating or implying relative importance or implying a quantity of indicated technical features. Therefore, a feature restricted by “first” or “second” may explicitly indicate or implicitly include at least one of such features. In the descriptions of the present disclosure, unless explicitly specified, “multiple” means at least two, for example, two or three.

[0101] In the present disclosure, it should be noted that unless otherwise explicitly specified and limited, the terms “mount”, “connect”, “connection”, and “fix” should be understood in a broad sense. For example, a connection may be a fixed connection, a detachable connection, or an integral connection; may be a mechanical connection or an electrical connection; and may be a direct connection, an indirect connection by an intermediary, or internal communication between two elements or mutual action relationship between two elements, unless otherwise specified explicitly. A person of ordinary skill in the art can understand meanings of the terms in the present disclosure based on certain situations.

[0102] Although the embodiments of the present disclosure have been shown and described above, it can be understood that, the foregoing embodiments are examples and should not be understood as limitation to the present disclosure. A person of ordinary skill in the art can make changes, modifications, replacements, or variations to the foregoing embodiments within the scope of the present disclosure.

What is claimed is:

1. An apparatus for battery dynamic equalization, comprising:

- a power battery comprising a first battery pack and a second battery pack connected to each other, and a first node disposed between the first battery pack and the second battery pack;
- a motor controller, a first end of the motor controller connected to a positive electrode of the first battery pack, and a second end of the motor controller connected to a negative electrode of the second battery pack;
- a motor, a first end of the motor connected to the motor controller, and a second end of the motor connected to the first node; and
- a controller connected to the motor controller, and configured to control, in a first state, a bridge arm of the motor controller to drive the motor to alternately charge and discharge the first battery pack and the second battery pack to self-heat the first battery pack and the second battery pack, and to configure an absolute value of a difference between a capacitance of the first battery pack and a capacitance of the second battery pack to be lower than a threshold.

2. The apparatus according to claim 1, wherein:

the motor controller comprises at least two first bridge arms, a first end of each of the at least two first bridge arms is connected to the positive electrode of the first battery pack, a second end of each of the at least two

first bridge arms is connected to the negative electrode of the second battery pack, and the at least two first bridge arms respectively correspond to at least two phases;

the motor comprises at least two first inductors, a first end of each of the at least two first inductors is connected to a corresponding first bridge arm, a second end of each of the at least two first inductors is connected to the first node, and the at least two first inductors respectively correspond to at least two phases; and

a neutral wire is led out from the motor, and is connected to the first node.

3. The apparatus according to claim 2, wherein the controller is configured to:

collect a battery equalization parameter, a temperature requirement parameter, and a driving requirement parameter of the power battery;

obtain a target driving current based on the driving requirement parameter, obtain a target self-heating current based on the temperature requirement parameter, and obtain a target equalizing current based on the battery equalization parameter; and

adjust, based on the target driving current, the target self-heating current, the target equalizing current, and an actual phase current, a first control signal for controlling the motor controller, to drive the motor, and self-heat and equalize the first battery pack and the second battery pack.

4. The apparatus according to claim 3, wherein the controller is configured to:

obtain a torque output of the motor, a rotational speed of the motor, a voltage of the power battery, and an electrical angle of a rotor of the motor; and

in response to that a driving instruction is received, obtain the target driving current based on the torque output of the motor, the rotational speed of the motor, and the voltage of the power battery, and obtain a differential mode voltage of each bridge arm based on the target driving current, the actual phase current, and the electrical angle of the rotor of the motor.

5. The apparatus according to claim 4, wherein the target driving current is a synthesis of a d-axis target current and a q-axis target current, the actual phase current is a synthesis of a d-axis actual current and a q-axis actual current, and the controller is configured to:

control the d-axis target current, the d-axis actual current, the q-axis target current, and the q-axis actual current, to obtain a d-axis target voltage and a q-axis target voltage; and

obtain the differential mode voltage of each bridge arm based on the d-axis target voltage, the q-axis target voltage, and the electrical angle of the rotor of the motor.

6. The apparatus according to claim 5, wherein the controller is further configured to:

obtain a target heating equalizing current based on the target self-heating current and the target equalizing current;

obtain an actual phase current of the motor;

obtain a common mode voltage based on the target heating equalizing current and the actual phase current; and

adjust the first control signal of the motor controller based on the differential mode voltage of each bridge arm and the common mode voltage.

7. The apparatus according to claim 3, wherein the controller is configured to:

collect the battery equalization parameter and the temperature requirement parameter of the power battery; and

obtain the target equalizing current based on the battery equalization parameter, and obtain the target self-heating current based on the temperature requirement parameter.

8. The apparatus according to claim 7, wherein the controller is configured to perform integral calculation on the battery equalization parameter to obtain the target equalizing current,

the battery equalization parameter comprises a discharge capacity difference, a voltage difference, or an instantaneous power difference, and

the discharge capacity difference, the voltage difference, and the instantaneous power difference respectively denote a discharge capacity difference, a voltage difference, and an instantaneous power difference between the first battery pack and the second battery pack.

9. The apparatus according to claim 8, wherein the battery equalization parameter is the discharge capacity difference, and

the controller is configured to: collect each phase current of the motor; calculate a neutral wire current of the motor based on each phase current of the motor; and integrate the neutral wire current to obtain the discharge capacity difference.

10. The apparatus according to claim 8, wherein the battery equalization parameter is the discharge capacity difference, and

the controller is configured to: collect a bus positive current and a bus negative current of the power battery; perform ampere-hour integration on the bus positive current of the power battery to obtain a discharge capacity of the first battery pack, and perform ampere-time integration on the bus negative current of the power battery to obtain a discharge capacity of the second battery pack; and obtain the discharge capacity difference based on the discharge capacity of the first battery pack and the discharge capacity of the second battery pack.

11. The apparatus according to claim 8, wherein the battery equalization parameter is the discharge capacity difference, and

the controller is configured to: collect a bus positive current of the motor controller and a bus negative current of the motor controller; perform ampere-hour integration on the bus positive current to obtain a discharge capacity of the first battery pack, and perform ampere-time integration on the bus negative current to obtain a discharge capacity of the second battery pack; and obtain the discharge capacity difference based on the discharge capacity of the first battery pack and the discharge capacity of the second battery pack.

12. The apparatus according to claim 8, wherein the battery equalization parameter is the voltage difference; and

the controller is configured to: collect a bus voltage of the motor controller and a voltage of the second battery pack; calculate a voltage of the first battery pack based on the bus voltage and the voltage of the second battery pack; and determine the voltage difference based on the voltage of the first battery pack and the voltage of the second battery pack.

13. The apparatus according to claim 8, wherein the battery equalization parameter is the instantaneous power difference, and

the controller is configured to: collect a bus positive current and a bus negative current of the power battery, a voltage of the first battery pack, and a voltage of the second battery pack; determine instantaneous power of the first battery pack based on the bus positive current and the voltage of the first battery pack, and determine instantaneous power of the second battery pack based on the bus negative current and the voltage of the second battery pack; and determine the instantaneous power difference based on the instantaneous power of the first battery pack and the instantaneous power of the second battery pack.

14. The apparatus according to claim 7, wherein the controller is configured to obtain required heating power based on the temperature requirement parameter, determine an amplitude and a frequency of a self-heating current of the neutral wire based on the required heating power, and determine the target self-heating current based on the amplitude and the frequency.

15. The apparatus according to claim 14, wherein the controller is further configured to obtain a target heating equalizing current by:

$$In^* = indc + ipk \cdot \sin(2 \cdot \pi \cdot f \cdot t),$$

wherein In^* denotes the target heating equalizing current, $indc$ denotes the target equalizing current, ipk denotes an amplitude of the target self-heating current, f denotes a frequency of the target self-heating current, and t denotes a time.

16. The apparatus according to claim 1, wherein, in response to detecting that the absolute value of the difference between the capacitance of the first battery pack and the capacitance of the second battery pack is not lower than the threshold in a second state, the controller is configured to control the bridge arm of the motor controller to equalize the first battery pack and the second battery pack.

17. A control method for an apparatus for battery dynamic equalization, wherein

the apparatus comprises a power battery, a motor controller, a motor, and a controller, the power battery comprises a first battery pack and a second battery pack connected to each other, a first node is disposed between the first battery pack and the second battery pack, a first end of the motor controller is connected to a positive electrode of the first battery pack, a second end of the motor controller is connected to a negative electrode of the second battery pack, a first end of the motor is connected to the motor controller, a second end of the motor is connected to the first node, and the controller is connected to the motor controller, and

the method comprises:

controlling, in a first state, the motor controller to drive the motor, alternately charge and discharge the first battery pack and the second battery pack to self-heat the first battery pack and the second battery pack, and to configure an absolute value of a difference between a capacitance of the first battery pack and a capacitance of the second battery pack to be lower than a threshold.

18. A vehicle, comprising an apparatus for battery dynamic equalization, wherein the apparatus comprises:

a power battery comprising a first battery pack and a second battery pack connected to each other, and a first node disposed between the first battery pack and the second battery pack;
a motor controller, a first end of the motor controller connected to a positive electrode of the first battery

pack, and a second end of the motor controller connected to a negative electrode of the second battery pack;

a motor, a first end of the motor connected to the motor controller, and a second end of the motor connected to the first node; and

a controller connected to the motor controller, and configured to control, in a first state, a bridge arm of the motor controller to drive the motor to alternately charge and discharge the first battery pack and the second battery pack to self-heat the first battery pack and the second battery pack, and to configure an absolute value of a difference between a capacitance of the first battery pack and a capacitance of the second battery pack to be lower than a threshold.

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